

Calculations for Strength of electromagnets:

Assumptions:

- the chain will weigh 650g
- the electromagnetic coils will run @ 1.5A
- the area of the magnetic pole is 8mm
- 4 electromagnets

$$\mu_0 = 4\pi \cdot 10^{-7} \text{ H/m}$$

- the permanent magnet offers no assist

How many turns of wire do we need?

$$B = \frac{\mu_0 N I}{g}$$

$$F \approx \frac{B^2 A}{2\mu_0}$$

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$$\frac{2g^2 F}{\mu_0 I^2 A} = N^2$$

$$N = g \sqrt{\frac{2F}{\mu_0 I^2 A}}$$

$$F = \frac{W}{4}$$

$$= \frac{1.6}{4}$$

$$F = 1.59 \text{ N}$$

$$N = 600 \text{ turns}$$

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F = desired force in Newtons

N = # of coil turns

I = coil current in Amperes

A = pole face area in m²

g = air gap in meters

G = gravity = 9.81 m/s²

They will thus even account for efficiency losses 600 turns is an upper end and a conservative first design.

Now lets calculate heat:

Resistance of 28 Awg wire $\approx 0.213 \Omega/\text{m}$

28 Awg insulated diameter $\approx 0.35 \text{ mm}$

Core diameter: 8mm

600 turns

Length of Core: 20mm

Turns per layer:

$$\frac{20}{0.35} \approx 57 \text{ turns/layer}$$

$$\# \text{ of layers} = \frac{600}{57} \approx 10.5 \approx 11$$

$$\# \text{ 1st layer} = 8 + 0.35 = 8.35 \text{ mm}$$

$$\text{element layer} = 8 + 2(0.35) = 8.7 \text{ mm}$$

$$\text{average diameter} \approx \frac{8.35 + 8.7}{2} = 8.5 \text{ mm}$$

$$\text{average length per turn} \approx \pi(1.185) \approx 3.72$$

$$\text{Total wire length} = 600 \times 3.72$$

$$\approx 2.23 \text{ m}$$

$$\text{Resistance} \approx (0.213 \Omega/\text{m} \times 2.23 \text{ m})$$

$$\approx 4.75 \Omega$$

$$P = I^2 R$$

$$\approx (1.5)^2 \times 4.75 \Omega$$

$$\approx 10.69 \text{ W of energy released as heat}$$

This will likely be far too much heat for the electromagnet to handle

Problem Solving: Where can we improve thermal efficiency?

$$P = I^2 R$$

If we can decrease I, or the cost of R (having big more turns) or reducing R.

- Permanent magnets onboard to reduce load

- Larger core

- More windings